

Minidiss – Group Meeting 1

24.09.2025

Part 1: Teamwork set-up

Make name cards.

Fill in **teamwork survey**: <https://forms.office.com/e/ckPJVt6kfA>



Introductions

- Name?
- Where are you from?
- Why do you study psychology?

Part 2: Topic intro

Today’s aim: “Introduce topic and major theme of the literature review, highlight key papers”

The topic: Statistical learning

If you woke up in a brand new place, surrounded by people using a language you don’t know at all, how would you figure out what the words in this unknown language are? How would your mind figure out where one word ends and the next one begins?

As you experience this language more and more, you’ll eventually develop a sense that, if you hear a certain syllable, you can predict what syllable comes next. But for other syllables, you might never be able to guess what comes next. Think about the English sequence “pretty baby”, which sounds something like “pri-tee bay-bee”. After you’ve heard “pri”, there aren’t that many possible syllables that can follow in English, besides “tee”. If you hear “pri”, you can be fairly confident that what’s coming next is “tee”. But after you’ve heard “pri-tee”, lots of different syllables could follow: if it’s “pretty baby”, you have “tee-bay”; if it’s a “pretty dog”, you have “tee-dog”, if it’s a “pretty sunrise”, you have “tee-sun”...

In general: when you hear syllable 1 and can predict what syllable 2 will be, those syllables tend to be the same word. But when you hear syllable 1 and aren’t certain at all about what syllable 2 will be, those syllables are probably in separate words.

In technical terms, we talk about the “transitional probability” (TP) from one syllable to the next. When transitional probability between syllables is high, you can be fairly confident about what’s coming next—and when TP is high, the syllables tend to be in the same word. When transitional probability between syllables is low, you don’t know at all what’s coming next—and when TP is low, the syllables tend to be in two different words.

The process of paying attention to transitional probabilities, and using them to learn words (or non-linguistic sequences), is called “statistical learning”. In this project, we’ll take the general topic of statistical learning in one of two possible directions, depending on your interests.

Possible directions for our project

Neurodiversity

Autistic people are said to focus more on smaller details of their input, compared to allistic (that is, non-autistic) people. Do autistic people tend to pay attention to subtler statistical cues in their input than allistic people do? Would autistic people tend to segment their input into smaller units than neurotypical people?

Visual perception

In signed languages, the continuous stream of input is not auditory, but visual. Previous research on visual statistical learning has focused on sequences of static shapes, but gestures are dynamic and linear across time, much like syllables of spoken language. Does statistical learning of gesture sequences pattern more like visual statistical learning, because it’s in the visual modality, or auditory statistical learning, because the units are dynamic over time?

Key literature

1. Saffran, J. R., Newport, E. L., & Aslin, R. N. (1996). Word segmentation: The role of distributional cues. *Journal of Memory and Language*, 35(4), 606–621. <https://doi.org/10.1006/jmla.1996.0032>
2. Siegelman, N., Bogaerts, L., & Frost, R. (2017). Measuring individual differences in statistical learning: Current pitfalls and possible solutions. *Behavior Research Methods*, 49(2), 418–432. <https://doi.org/10.3758/s13428-016-0719-z>
3. Emberson, L. L., Conway, C. M., & Christiansen, M. H. (2011). Timing is everything: Changes in presentation rate have opposite effects on auditory and visual implicit statistical learning. *Quarterly Journal of Experimental Psychology*, 64(5), 1021–1040. <https://doi.org/10.1080/17470218.2010.538972>

Part 3: Prep for our next group meeting

In our next meeting in Week 5, we’ll choose a direction for our project, and dive deeper into the literature.

What I’ll do

I’ll compile the results of the teamwork survey and distribute the data to all of you. This information should help you delegate tasks and find good times to collaborate.

What you’ll do

Individually

- **Read the key papers from today.** (Don’t worry too much about the fine details of the experiment/analysis/etc. Focus on understanding the overall ideas and the authors’ argumentation in the introduction, discussion, and conclusion.)
- **Think about what research tasks you might like to spearhead / share responsibility for, based on your skills.** (Everyone will ideally do a bit of everything, but let’s distribute the load a little.) Example tasks:
 - Reading and synthesising previous literature
 - Drafting the ethics application
 - Creating the experiment
 - Analysing the data
 - Designing the poster
 - Presenting the poster
 - Other?

As a group

1. **Reach an agreement with one another about which two or three people want to focus on which different elements of the research process, based on your strengths and skills.** (Again, everyone will ideally do a little bit of everything, but these people will be responsible for making sure each step gets done.)
2. After everyone’s read the key papers, discuss which research direction interests you more: **statistical learning and neurodiversity** or **visual perception of gesture sequences**. Take a vote and show me the results next time we meet.

If you need me

Email me at elizabeth.pankratz@ed.ac.uk

Or book office hours with me at <https://edin.ac/3K9TFC3>